

Remarkable Franz-Keldysh Effect in Ge-on-Si p-i-n Diodes *

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Ge-on-Si p-i-n diodes are fabricated by using two-step Ge film epitaxial technology on Si substrates. A remarkable Franz-Keldysh effect is observed in the wavelength range of 1620–1640 nm with a largest $\Delta\alpha/\alpha$ of 2.8 at 1640 nm by optical responsivity measurement. The remarkable change of absorption coefficient in the considerable large wavelength range makes Ge-on-silicon a promising candidate for Si-based electro-absorption modulators. The initial design predicts a modulator of bandwidth ~ 50 GHz, and the extinction ratio > 7 dB by the measured parameter.

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It is hoped that optical technology will be revolutionary for on-chip interconnects. The most promising candidate is silicon photonics. In this field, the optical modulator is one of the workhorses of such interconnection and has proved to be difficult to realize in practical devices.^[1–3] In the past ten years optical modulators have dramatically improved with a notable increase in the bandwidth from 100 s MHz to 30 GHz.^[4–16] However, due to the weak electro-optical effect in silicon, these kinds of modulators mostly utilize the plasma dispersion effect. At present, however, because of the weak free-carrier effect, it is difficult to find the correct balance of extinction ratio (ER), bandwidth and optical bandwidth for silicon modulators based on that effect. Moreover, they mostly have large power consumption.^[10–16] Compared with the modulators mentioned above, electro-absorption (EA) modulators could potentially achieve low operating energy, large bandwidth and practical optical bandwidth, because EA effects do not rely on carrier density change. Given the weak electro-optical effect of silicon and the advantage of the EA modulator, some researchers have tried to introduce a second material to realize a silicon-based EA modulator. However, this material must have at least two features: (1) a large EA effect to apply to silicon-based EA modulator; (2) compatibility with CMOS technology. Germanium completely satisfies those two requirements. Firstly, germanium has a direct band gap of 0.8 eV, corresponding to the on-chip communication wavelength of 1.55 μm . This gap can be fine tuned by introducing silicon. At close to 1.55 μm , germanium behaves essentially like direct gap materials such as GaAs, and is expected to have the relevant remarkable optical effect. Secondly, Ge

is completely compatible with silicon technology. In this Letter, the Franz-Keldysh (FK) effect of Ge films on Si substrates is discussed. The results indicate that a notable change of the Ge absorption coefficient $\Delta\alpha/\alpha \sim 2.8$ in the wavelength range of 1620–1640 nm is observed under an electric field of 62.5 kV/cm. Based on the results, we choose an optimal working region suitable for an EA modulator. Because the bandwidth of this kind of EA modulator based on the FK effect is mainly dominated by the resistor and the capacitance, we predict that the initial modulator can potentially achieve a bandwidth ~ 50 GHz and meanwhile can retain $\text{ER} > 7$ dB.

Figure 1 shows the schematic cross section view of the Ge p-i-n diodes used in this work to investigate the remarkable EA effect of Ge in a C band. The device was fabricated on an N^+ -doped Si substrate. Before the Ge epitaxy, the substrate was degassed at 300°C in the pretreatment chamber and deoxidized at 930°C for 5 min. A 60 nm thin buffer Ge layer was grown on Si using cold-wall UHV/CVD at low temperature 300°C followed by a 800-nm-thick Ge layer at high temperature 580°C. The double mesa of the diode was fabricated by standard lithography, dry etching, and metallization processes, and the diameter was 25 μm . The top Ge layer was implanted with BF_3^+ at a dose of $4 \times 10^{15} \text{ cm}^{-2}$ to form the p-type contact. Especially it is worth noticing that a 640-nm-thick SiO_2 film and a 490-nm-thick Si_3N_4 film were finally deposited as the passivation layer and optimized in terms of thickness in order to form an anti-reflection layer with the transmission rate of 90% to improve the diode's responsive characteristic to obtain a relatively large optical current.

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The FK effect of the device was studied by measuring the optical current for various reverse voltages. To make sure that the FK effect is the only factor resulting in the optical current change, we measured the C – V characteristics. Figure 2 illustrates the test result. When the reverse voltage is smaller than -0.61 V, the capacitance tends to a stable value 10.28 pF, which means that the Ge layer tends to be completely depleted. It also proves that the unintended doping is very low. Correspondingly, the electric field in i-Ge is uniform for the $V_{\text{reverse}} < -0.61$ V. If we only measure and compare the optical current at $V_{\text{reverse}} < -1$ V, we can arguably rule out the influence of the variation of the p–n junction's space charged region with respect to reverse bias on the photocurrent change.

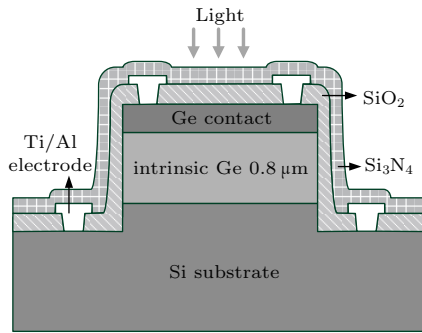


Fig. 1. Schematic cross-section structure of the device.

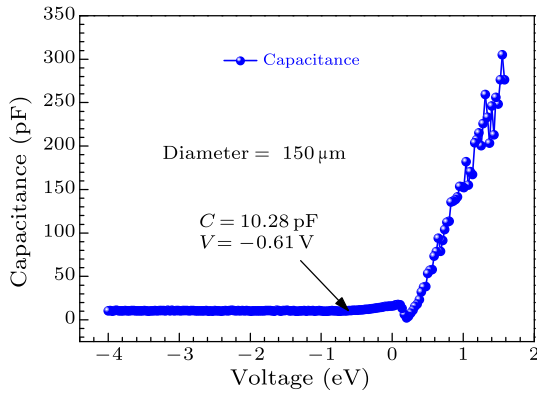


Fig. 2. Capacitance of the diode at different reverse voltages.

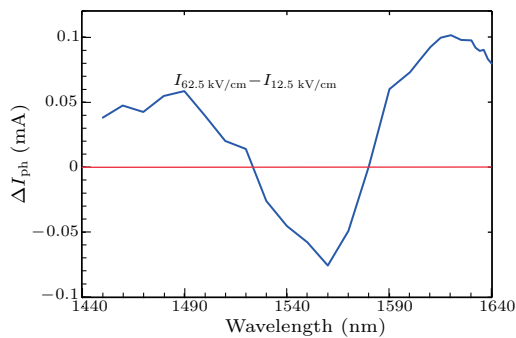


Fig. 3. The difference of the photocurrent at 5 V and 1 V, showing the characteristics of the FK effect.

The optical current of the diode was measured at different reverse voltages from 1 V to 5 V with steps of 0.05 V (corresponding to 12.5 kV/cm and 62.5 kV/cm). The incident light power was 4.8 mW and was kept to be constant during the measurement. Figure 3 shows the photocurrent difference ΔI between 1 V and 5 V.

The photocurrent spectrum shows the characteristics of the FK effect, which includes a tail at energies below the band gap and oscillations above it.^[17–19] The tail indicates that when an external electric field is applied on the body material, the absorption of the photon energy below the bandgap will remarkably increase. This electric field-induced absorption change is the physical basis for the EA modulator on silicon.

The absorption coefficient can be calculated from the photocurrent at a certain reverse voltage for a specific wavelength using the formula

$$\alpha = -\frac{1}{t_{\text{Ge}}} \ln \left(1 - \frac{1240}{\lambda(\text{nm})} \frac{I_{\text{ph}}}{P_{\text{out}}(1 - R_{\text{refl}})} \right),$$

where I_{ph} is the photocurrent, P_{out} is the normal incidence light power out from an uncovered fiber, and R_{refl} is the reflectance of the photodiodes. In the treatment, the contribution of the indirect band gap to the optical current was ignored. Figure 4 shows the computed absorption coefficient using the formula at 1 V and 5 V, separately.

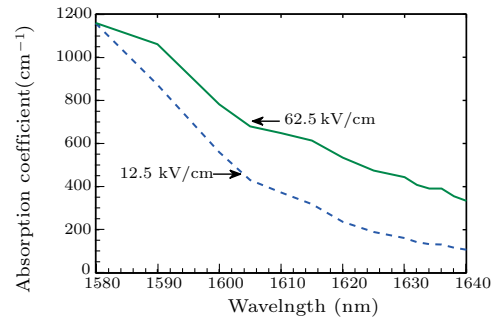


Fig. 4. The calculated absorption coefficient under electric field of 12.5 V/cm and 62.5 V/cm, respectively.

The electric induced absorption coefficient change is relatively remarkable in the wavelength range 1605 – 1640 nm. The largest change of absorption coefficient was observed at 1625 nm from 534 to 297.2 cm $^{-1}$. However, because of the high “off-state” absorption, using this wavelength as working region will introduce a large insert loss (IL). Usually $\Delta\alpha/\alpha$ is used as a figure of merit to characterize a modulator and to choose out the optimal working region. Our measurement achieved the largest figure value about 2.8 at 1640 nm and $\Delta\alpha/\alpha > 2$ for 1620 – 1640 nm, which means a large tunable region compared with the micro-ring modulators.

Using the measured data we calculate and choose the optimal working region for EA modulators based on the FK effect. The active length of the modulator

is 70 μm , and the height and width are 500 nm and 400 nm, respectively. Because the bandwidth of this kind of the EA modulator based on the FK effect is mainly dominated by the resistor and the capacitance, the 3 dB bandwidth is expected to be ~ 50 GHz.

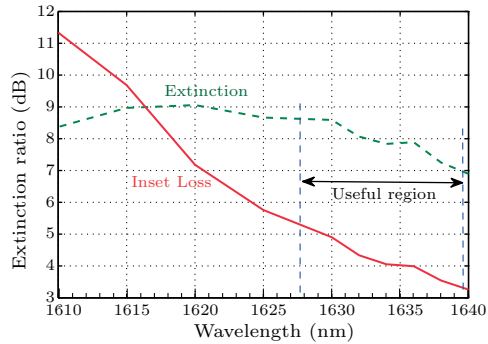


Fig. 5. The useful working region for a 70- μm -long modulator. The wavelength range is 1627–1640 nm with $\text{ER} > 7$ dB and $\text{IL} < 5$ dB at any given wavelength.

The ER and IL are shown in Fig. 5. An ER of > 7 dB is achieved in the wavelength range 1610–1640 nm. The maximum ER is 9 dB at 1620 nm. This value can be further increased by increasing the active region, however, accompanied by increasing IL. Both the ER and IL determine the operating spectrum range of the device. In this calculation, the most optimal operating spectrum range is 1627–1640 nm, which is 13 nm wide covering nearly half of the C band. Across the whole waveband the ER is > 7 dB and the ER is < 5 dB everywhere. We think that this wavelength range completely satisfies the requirement of an EA modulator. This working region can be shifted to the C band by doping Si to Ge, where the content of Si is about 1.28% by the band calculation.

In conclusion, a vertical p-i-n Ge diode is achieved

successfully by using the low- and high-temperature epitaxy method by UHV-CVD to achieve a good-quality Ge material on the Si substrate. In this diode we observe a remarkable FK effect in the wavelength range of 1620–1640 nm with a large absorption coefficient change $\Delta\alpha/\alpha = 2\text{--}2.8$. We also calculate and fix a optimal working spectrum region for a 70- μm -long practical modulator with adequate ER and low IL, and the 3 dB bandwidth is expected to be ~ 50 GHz. In the future we will further precisely design and fabricate this kind of waveguide integrated modulators based on this remarkable Ge-on-Si FK effect using the present experimental and calculated data.

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